Response to Review: JCGS-25-420.R1

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We thank the anonymous associate editor and reviewers for their careful reviews. What follows is our point-by-point response to reviewer comments. The original reviewer comment is in normal font and *our response is in italic text.*

This revision and the comments given by the authors do not successfully address most of my previous major and minor concerns. My impression is that they attempted a "quick fix" by adding some new material here and there without substantially reworking the narrative, evaluation, and presentation quality of the manuscript and did not engage with the substance of my previous points in good faith. Despite my criticisms, I do think the core idea of HBE + tour overlays is very interesting and innovative and has large potential value as a diagnostic tool for NLDR embeddings. However, in its current form, the manuscript is not suitable for publication in my opinion.

Prior Major Points

1) Missing comparisons to standard embedding-quality measures (co-ranking / RNX, trustworthiness/continuity, stress, Shepard diagrams, etc.)

Partly addressed. The revision now explicitly discusses several established metrics in Section 3.4, and it includes numerical/visual comparisons of HBE against metrics such as "rARNX", "rRTA", "rGS", and a Shepard-diagram-based correlation (e.g., Figure 8 and the surrounding discussion). However, the new material mostly *illustrates that metrics can disagree rather than establishing when/why HBE should be preferred* or how an analyst should reconcile HBE with existing (and complementary) local-vs-global criteria. The revised narrative is too dismissive of disagreement between metrics as "confusion" rather than treating it as an unavoidable multi-objective trade-off (see below).

We thank the reviewer for this thoughtful and constructive comment. We appreciate the recognition that the revised manuscript includes a broader set of comparisons to established *embedding-quality measures. However, we agree that the current presentation does not yet sufficiently clarify when and why HBE should be used, nor how it relates to existing metrics in a principled way.*

In particular, we acknowledge the reviewer's point that our previous wording may have been overly dismissive in describing disagreements between metrics as "confusion." We agree that such differences are more appropriately understood as an inherent feature of NLDR, which involves multiple competing objectives, such as preserving local neighborhoods versus global structure.

2) Overselling the method as a decision aid for choosing "the best" 2-D embedding vs positioning it primarily as a diagnostic/exploratory tool

Not addressed (and arguably worse). The paper still contains a full section titled *"Choosing the best 2-D layout"* (Section 4), and the surrounding prose continues to frame the workflow as "select the most accurate representation" rather than "diagnose and understand distortions and failure modes of a chosen representation." In their response letter, the authors themselves explicitly stated that

"HBE is intended as a diagnostic tool rather than a definitive selection metric" but the submitted manuscript text does not reflect this position at all.

We thank the reviewer for highlighting this important issue. We fully agree that the manuscript, in its previous form, continued to frame the method as a tool for selecting a "best" 2-D layout, which is not consistent with our intended contribution.

As the reviewer correctly notes, NLDR involves inherent trade-offs between competing objectives (e.g., local versus global structure preservation), and therefore the notion of a single universally "best" embedding is generally not well-defined. We also acknowledge the inconsistency between our response letter—where we described HBE as a diagnostic tool and the manuscript text, which continued to use selection-oriented language.

3) Section 3.5 (out-of-sample / prediction into an embedding)

Only very partially addressed. Per the response letter, citations to related work have been added and the authors frame this as future work. But the manuscript still gives it prominent space in the methods section; there is no theoretical argument, no empirical validation, and no comparison to existing out-of-sample embedding approaches.

We agree that, in the previous revision, Section 3.5 gave the impression of a fully developed out-of-sample embedding method without sufficient theoretical justification, empirical validation, or comparison to existing approaches. This was not our intention.

In response, we have revised Section 3.5 in several ways:

- *Repositioning of the contribution: The section has been rewritten to clearly present the proposed approach as a simple, model-consistent approximation based on the fitted wireframe, rather than as an alternative to established out-of-sample embedding methods. We explicitly state that it is intended for exploratory use.*
- *Reduced emphasis: The prominence of this section has been reduced, and the language has been made more cautious to avoid overstating its role within the overall methodology.*
- *Initial empirical check: We have added a basic comparison with the UMAP `predict()` function using a train/test split. This provides a preliminary assessment of how the approach performs in terms of neighborhood preservation and consistency with the fitted model.*
- *Clarification of limitations: We now explicitly acknowledge that this is only an initial investigation, and that a more comprehensive evaluation across methods and datasets remains future work.*

These changes aim to ensure that the section is aligned with the primary goal of the paper; developing diagnostic tools for understanding NLDR layouts, while avoiding overstating the capabilities of this component.

4) Section 3.6 (parameter tuning guidance)

Not addressed in substance. The section is longer, but still reads like: "Try many settings, draw many plots, then pick something that looks OK." I still do not see clear, reproducible guidance that an analyst can follow to choose binwidth etc parameters in a systematic way. If none can be provided, this should be clearly stated and the method reframed as inherently exploratory/diagnostic rather than selection-oriented.

5) Writing/proofreading and overall presentation quality

Not addressed. Despite the claim in the response letter that the manuscript was carefully proofread, the revision still contains a distracting density of typos, malformed sentences, in consistent notation, and colloquial phrasing (examples below from Section 3.4 alone). The writing quality is still far below what I would expect for a publication-ready manuscript in JCGS.

Logistical aspects of the revision

The revision process is made unnecessarily difficult because none of the changes are highlighted and the response letter does not point to precise locations (page/line) nor quotes key replacements. I find this rather inconsiderate and fairly unprofessional. A tracked-changes PDF (or a detailed change log keyed to reviewer comments) would substantially reduce friction for reviewers and editors if another revision is submitted.

Summary

IMO, the manuscript still has at least two critical problems:

1. Claim–evidence mismatch: It is still written as if it provides a reasonably objective procedure for *choosing the best 2-D layout*, but the evaluation is not strong enough to support that framing.

2. Presentation quality: The prose is sloppy enough (typos, malformed sentences, non standard/unmotivated notation) that it materially impairs readability and confidence.

Major Issues

1. Still oversells "Choosing the best 2-D layout"

Section 4 is still titled **"Choosing the best 2-D layout"**, and the text repeatedly suggests that the goal is to select an "accurate" representation over a pretty but wrong representation. This framing remains problematic for the basic reason I raised previously: in 2-D NLDR, *there is generally no single optimum* – there is a trade-off between local and global structure preservation, and "best" is always implicitly "best for some downstream purpose."

The revision now acknowledges that metrics disagree because they emphasize different aspects of structure, but it still presents disagreement as something that "causes confusion" for the analyst, rather than as an expected feature of a multi-objective problem with objectives that are often in conflict.

I would suggest to rewrite Section 4 so it explicitly adopts the diagnostic framing stated in the response letter (identify local distortions + misleading global separations; *support comparative diagnosis*). If authors insist on "selection", they need to clearly define a *decision* rule and properly evaluate it (see next issue).

2. Validation remains insufficient for a selection claim (even with new metric comparisons)

The added comparisons to rARNX/rRTA/rGS/rSC are useful, but they do not resolve the central question: **Does HBE reliably help choose better embeddings, and in what sense of "better"?**

Right now, the paper still reads like a sequence of worked examples. The new metric com parison section mainly shows that different metrics rank layouts differently and then argues (qualitatively) that some metrics overemphasize global separation. That may well be true in specific clustering-heavy scenarios, but it does not constitute validation of HBE as a general purpose criterion.

In particular, if the paper's main claim is **diagnostics/exploration**, then the evidence should demonstrate that HBE + the tour overlay helps analysts localize + interpret distortions that are hard to see otherwise. (This seems to be there, already, but could be sharpened). If the paper's claim is **selection**, then the evidence given in the paper must demonstrate that a well specified selection procedure based on HBE selects embeddings that are "better" according to well-defined criteria across a range of controlled scenarios. (This is currently missing and, IMO, would almost necessarily require a simulation study with data with various known high dimensional structures as ground truth.)

We thank the reviewer for this careful and insightful comment. We agree that the current manuscript does not provide sufficient evidence to support HBE as a general-purpose selection criterion for choosing an optimal embedding, and we appreciate the reviewer's clear distinction between diagnostic and selection-oriented claims.

Our intention has always been closer to the former—namely, to provide a diagnostic and exploratory framework for understanding how an embedding represents the original high-dimensional data. We acknowledge that the manuscript, as previously written, did not consistently reflect this intention and instead suggested a stronger selection-oriented interpretation than is warranted by the current evidence.

We also appreciate the reviewer's suggestion that a selection claim would require a well-defined decision rule and systematic validation across controlled scenarios with known ground truth. We agree that such an evaluation is beyond the scope of the current work.

3. Section 3.6 still does not give reproducible, non-ad-hoc tuning guidance

With all due respect, my takeaway from the expanded Section 3.6 is still: "You can try a lot of things and draw some graphs and then maybe pick one of these settings, but who knows…" I might be missing something, but I do not see a clear recipe that an analyst could follow without substantial subjective judgment calls. If no such recipe can be given, this should be clearly stated and the method reframed as inherently exploratory/diagnostic rather than selection-oriented.

4. Section 3.5 (out-of-sample embedding) still reads like an unvalidated method

The response letter says this is "future work," and some citations were added. That is helpful, but it certainly cannot be presented as if it were an alternative to established projection or out-of-sample embedding methods just by claiming that it is. Without at least a basic empirical check (e.g., does it preserve neighborhood structure for held-out points as well as established out-of-sample methods?), this section should not be presented as a usable alternative.

5. Writing, proofreading, and notation quality are still far below publication-ready

The revised paper seems fairly sloppy and badly written overall. It was obviously not proofread or edited with any amount of proper diligence. There are too many typos, inappropriate colloquialisms, weird/bad phrasings, and non-sequiturs to list.

We sincerely apologize for the poor writing quality and notation issues that remained in the previous revision.

Just taking Section 3.4 as an example:

- **p. 12**: "ARNX" is not the abbreviation used in the cited work as far as I remember (and in any case should be carefully defined and kept consistent with the literature).

*The **ARNX notation has been corrected and aligned with standard usage in the literature**, and all metric names and abbreviations are now clearly defined and used consistently throughout the manuscript.*

- **p. 13, l. 6**: "The Shepard diagram (Shepard 1962) and its associated Spearman correlation (SC) (Spearman 1961) between p-D and k-D distances." – not a sentence (no verb/object).

Sentence fragments and malformed constructions (such as the Shepard diagram description noted on p. 13) have been rewritten as complete, grammatically correct sentences.

- **p. 13**: "model here is similar to a confirmatory factor analysis model (see a general explanation in Brown (2015)), $T+\hat{E}$" – the notation $T+\hat{E}$ is not introduced nor explained anywhere, and not used again anywhere else either; it just confuses/distracts.

*Unintroduced, unused, or confusing notation (e.g., expressions such as $T + \hat{E}$ and $\hat{X}$) has been **removed or properly defined**, and notation is now introduced only when it is subsequently used.*

- **p. 13, l. 31**: "our approach is similar to the ones used in this area. it is based on" – capitalization error.

Capitalization, punctuation, and phrasing errors have been corrected throughout.

- **p. 13**: "these fitted values might also be denoted as $\hat{X}$" – notation introduced and then never used again anywhere throughout the text.

- **p. 14**: “which occurs due the intentional design that one cluster is slightly 3-D and perfectly captured by a 2-D layout” – not valid English, and also plausibly inverts the intended meaning. I assume this should be something like: “which occurs due to the intentional design that this cluster is slightly 3-D and thus not able to be perfectly represented/preserved by a 2-D layout.”

Ambiguous or incorrect descriptions (such as the discussion of the 3-D cluster representation) have been rewritten to accurately reflect the intended meaning.

A similar density of errors can be found in most other sections. To me, this still seems far from publication-ready in terms of writing quality (scientific merit and remaining open points of discussion notwithstanding).

We recognize that the density of these issues in the previous version understandably undermined readability and confidence in the work. We are sorry for this, and we have made a concerted effort to ensure that the current revision meets the standards of clarity, consistency, and professionalism expected for a JCGS submission.